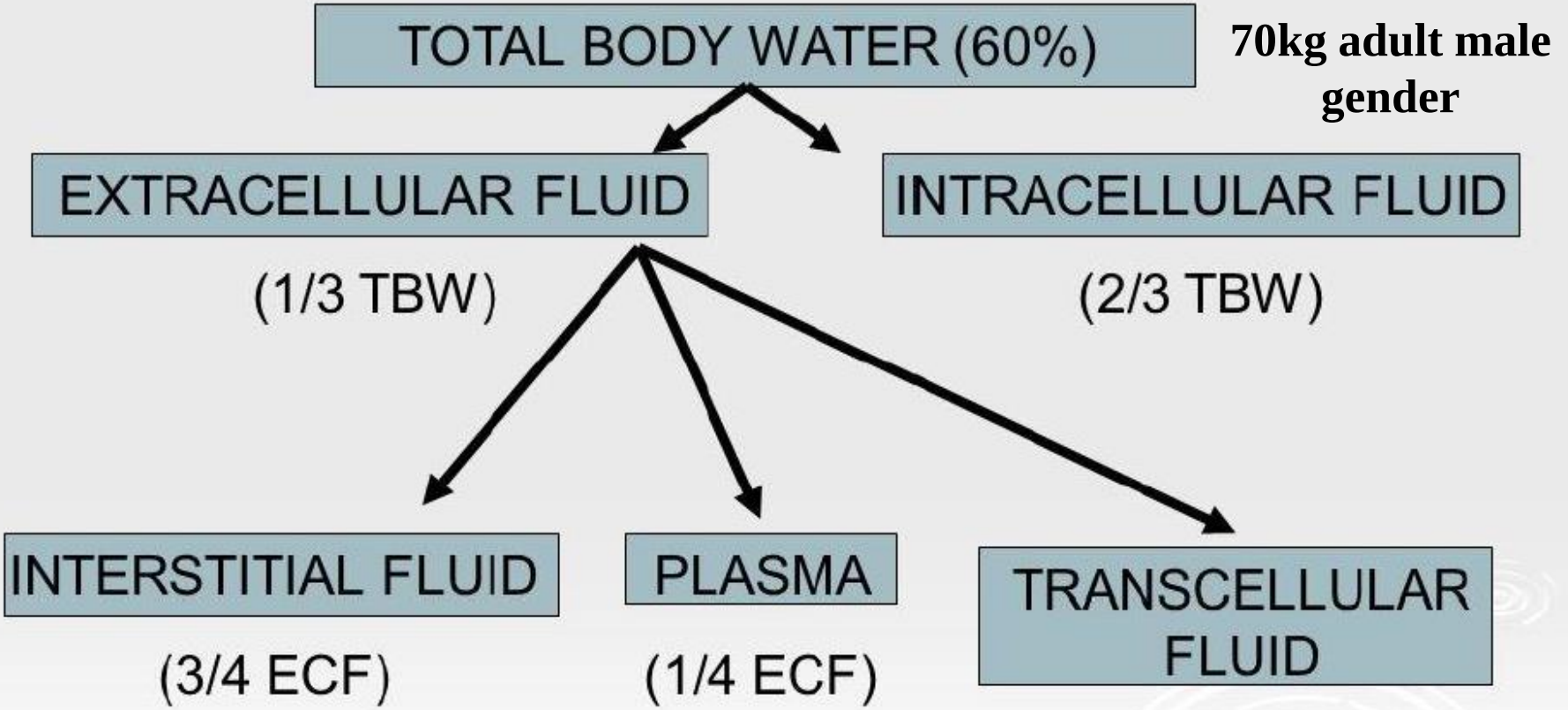
Disorders of salt balance: content

- Disturbance of volume and osmotic equilibrium
  - Body fluid components
  - Major electrolytes in body fluid
  - Cellular membrane permeability
  - Osmolality, effective osmolality and volume
  - Regulation: ANP, ADH and RAAS
  - Examples:
    - Hyperaldosteronism and hypoaldosteronism,
    - Diabetes insipidus and SIADH (Syndrome of Inappropriate Antidiuretic Hormone)

- Hypovolemic and hypervolemic states

- Review: Structure and function of nephron

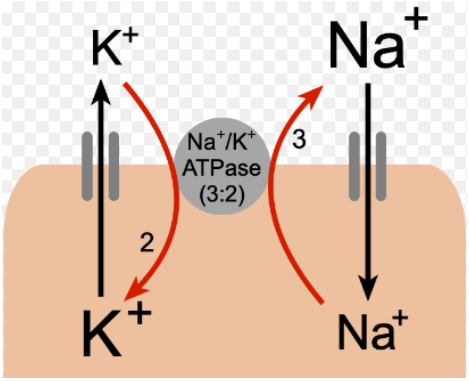
Body fluid components



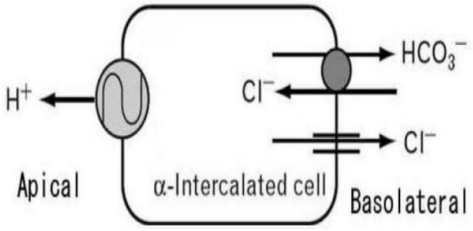
joint spaces, the pericardial sac, the peritoneal cavity,
And the pleural cavity: it is often referred to as third-space fluid.
total body water decrease with advancing age due to more adipose tissue and less muscle

Approximate major electrolyte content in body fluid

Extracellular fluid: plasma		Intracellular fluid	
Electrolytes	mmol/L	Electrolytes	mmol/L
Cations		Cations	
Na^+	135-145	K^+	150
K^+	3.5-5.5	Mg^+	40
Ca^+	5	Na^+	10
Mg^+	2		
Anions		Anions	
Cl^-	103	Phosphates and sulfates	150
HCO_3^-	26	HCO_3^-	10
HPO_4^{2-}	2	Proteinate	40
HSO_4^-	1		
Organic acids	5		
Proteinate	17		

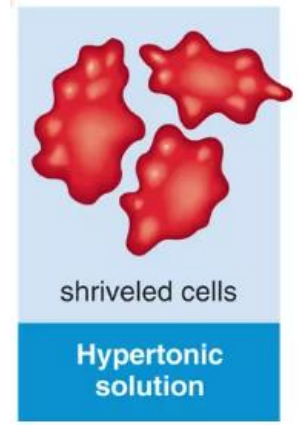


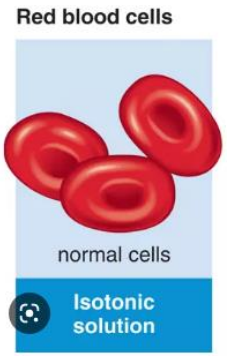
3Na+/2K+ ATPase pump

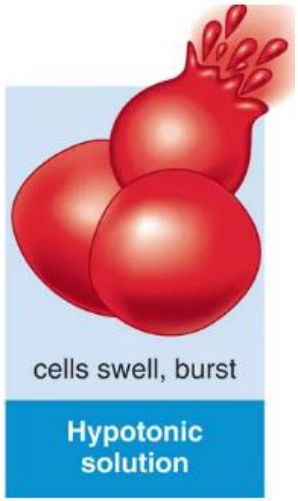


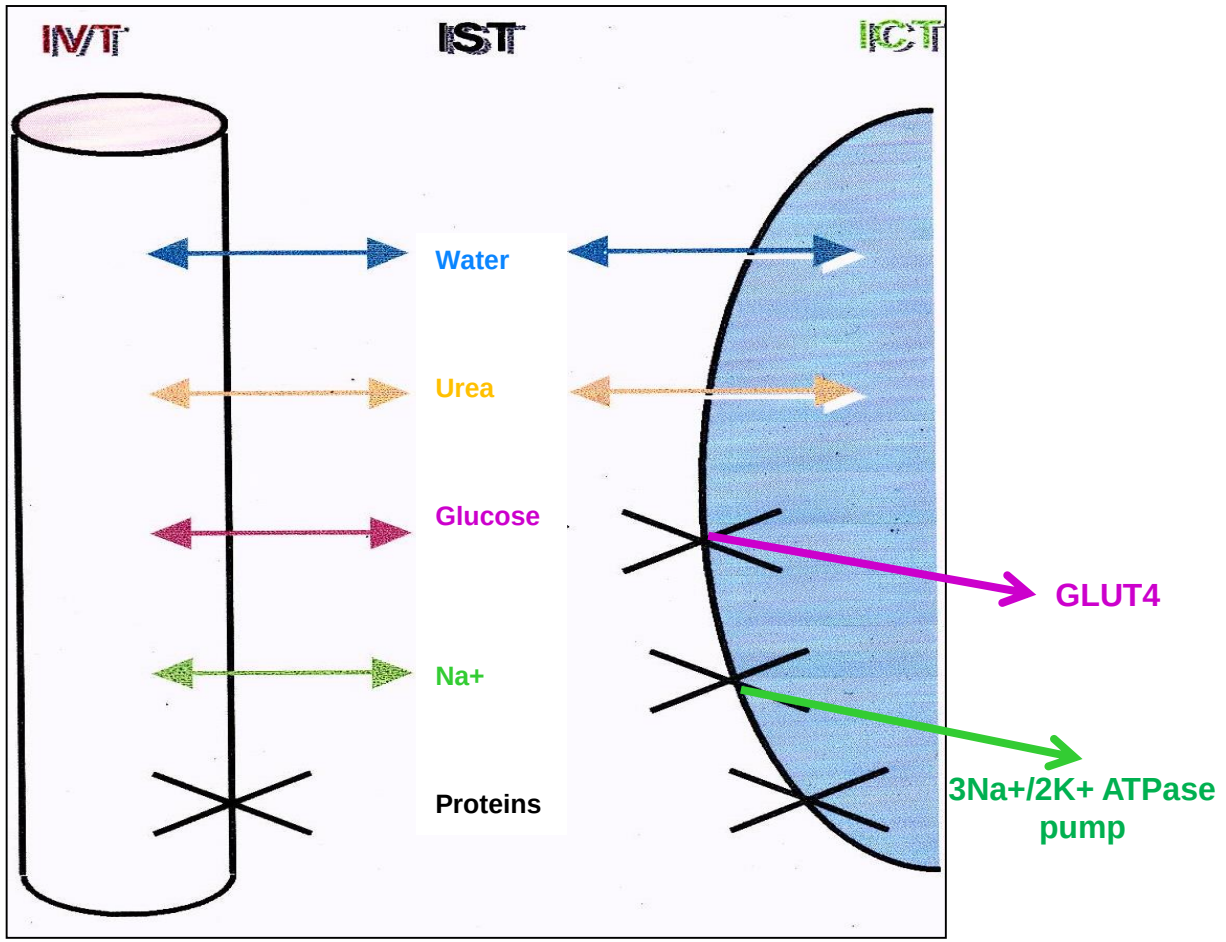
Cellular membrane permeability

The high permeability of the small pore system to electrolytes (except in the brain) quickly establishes an electrolyte equilibrium between plasma and interstitium. Thus it is only the plasma proteins, that exert a sustained osmotic pressure across the capillary wall.









capillary

cellular membrane

Osmolality

The concentration of osmotically active particles in solution expressed in osmols or milliosmols per kilogram of solvent.

- Normal osmolality range 285 to 295 mOsm/kg
  - Hypoosmolality: serum osmolality < 285 mOsm/kg
  - Hyperosmolality: serum osmolality > 295 mOsm/kg

- Osmolality (equation): $2Na^+$ (+ K^+) + glucose + urea (BUN: blood urea nitrogen)

- Osmotically active particles: Cl^-, HCO_3^-, alcohol, aceton and mannitol, proteins

Osmometers

- Osmometers are used for measuring osmolality of biological materials.
- Osmotic gap: the difference between calculated and measured osmolality



Osmolarity

The concentration of osmotically active particles in solution expressed in osmols or milliosmols per liter of solution.

Osmolality: Na^+ as a key player

- Na^+ affects osmolality to the larger extent than K^+, because Na^+ is the main extracellular cation (concentration)

- Normal osmolality range 285 to 295 mOsm/kg
  - Hypoosmolality: serum osmolality < 285 mOsm/kg
  - Hyperosmolality: serum osmolality > 295 mOsm/kg

- Osmolality (equation): $2Na^+$ + glucose + urea (BUN: blood urea nitrogen)
  - Na^+: 135-145 mmol/L
    - Mild (146-149 mmol/L)
    - Moderate (150-169 mmol/L)
    - Severe (≥170 mmol/L)
  - K^+: 3.5-5.5 mmol/L
  - Glucose: 3.3-5.5 mmol/l
  - Urea: 1.7–8.3 mmol/l
  - Albumine: 0.5-0.78 mmol/L (They constitute 55-65% of total plasma proteins)

Table 1. Severity Classifications Of Hypokalemia And Hyperkalemia

Condition	Mild (mEq/L)	Moderate (mEq/L)	Severe (mEq/L)
Hypokalemia	3.0-3.5	2.5-2.9	< 2.5
Hyperkalemia	5.5-6.4	6.5-7.5	> 7.5

Other osmolality equations

- Calculated osmolarity = 2 Na^+ + Glucose + Urea (all in mmol/L).
- or
- Calculated osmolarity = 2 Na^+ + 2 K^+ + Glucose + Urea (all in mmol/L).
- or
- Calculated osmolarity = 2[Na^+] + [Glucose]/18 + [BUN]/2.8 where [Glucose] and [BUN] are measured in mg/dL and [Na^+] in mmol/L. (typical in the US)
- or
- If the patient has ingested ethanol, the ethanol level should be included in the calculated osmolality.

EXAMPLE CASE

A sixty-seven year old white male was found pulseless and resuscitated; then brought to the emergency room. He had been reported to be drinking in a bar all afternoon, and had then fallen from a ten foot balcony to snow covered ground. He arrived in the emergency room with a fractured occiput and was unresponsive.

Na^+=143 mEq/l	BUN=4 mg/dL	pH=7.30
Cl^-=105 mEq/l	GLU=104 mg/dL	MO=356 mOsm/kg
HCO_3^-=19 mEq/l	KETO=Neg	

$AG = Na^+ - Cl^- + HCO_3^- = 143 - 105 - 19 = 19$
Normal AG = 8-16 mEq/L

$CO = 2 \times Na^+ + 1.15 * GLU/18 + BUN/2.8 = 286 + 1.15 * 104/18 + 4/32.8 = 294$

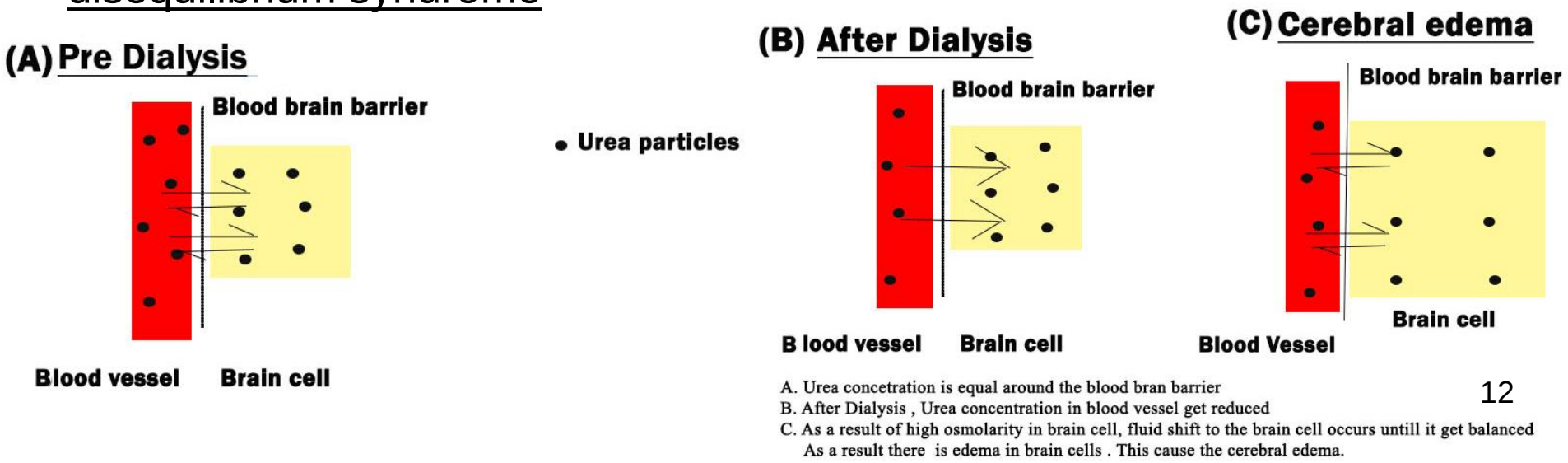
$0G = MO - CO = 356 - 294 = 62$

EXAMPLE CASE: Interpretation

- An osmolar gap value greater than 15 has traditionally been considered a critical value or cutoff. Approximately 97% of osmolar gaps in patients are between -10 and +10.
- Proper interpretation of the OG also requires knowledge of the anion gap (AG = Na^+ - HCO_3^- - Cl^-), the blood pH, and qualitative testing of the plasma ketone bodies (KETO)

Plasma tonicity, effective osmolality

- $2Na^+$ + glucose
- plasma tonicity reflects the concentration of solutes that do not easily cross cell membranes
- The most common example of this phenomenon is the rapid fall in plasma urea concentration produced by hemodialysis in a uremic patient. In this setting, the removal of plasma urea occurs more quickly than urea can equilibrate across the hemo-cerebral barrier (delayed in the brain due to hemo-cerebral barrier), which promotes osmotic water movement into the brain. In the brain, the water shift can result in cerebral edema and acute neurologic dysfunction, changes that partially explain the dialysis disequilibrium syndrome



Effective volume

- When the effective circulating blood volume is compromised, the condition is often referred to as *hypovolemia.*

- **Effective volume** — it is monitored by sensors in the renal afferent glomerular arterioles (for renin), carotid sinus (for antidiuretic hormone, ADH), and atria and ventricles (for natriuretic peptides) that respond to changes in pressure (baroreceptors), not volume.

- By contrast to the baroreceptors, osmoreceptor in hypothalamus are important for thirst sensation and to induce secretion of antidiuretic hormone

Hormonal factors: diuresis vs. water retention

Atrial natriuretic peptide (ANP) → **Diuresis**

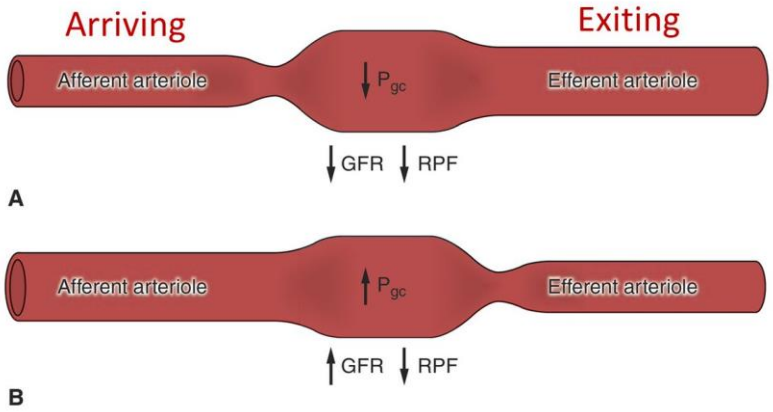
Renin-angiotensin-aldosterone + **Antidiuretic hormone** → **Water retention**

Atrial natriuretic peptide (ANP) or atrial natriuretic factor (ANF)

- A family of hormone/paracrine factors that are structurally related: ANP, BNP.
- BNP (brain-type natriuretic peptide) is synthesized largely by the ventricles as well as in the brain where it was first identified
- ANP is synthesized and secreted by cardiac muscle cells in the walls of the atria in the heart. These cells contain volume receptors (low pressure baroreceptors) which respond to increased stretching of the atrial wall due to increased atrial blood volume.
- Aims:
  - Suppress sympathetic outflow
  - Promote myocardial relaxation and inhibit hypotrophy and fibrosis
  - The main function of ANP is causing a reduction in expanded extracellular fluid (ECF) volume by
    - inhibition of secretion of renin and aldosterone
    - vasodilation of the afferent arterioles with constriction of the efferent arterioles leading to increased GF and decreased sodium and water absorption by kidney tubules.

The model of glomerular filtration rate (GFR) and renal perfusion (RPF)

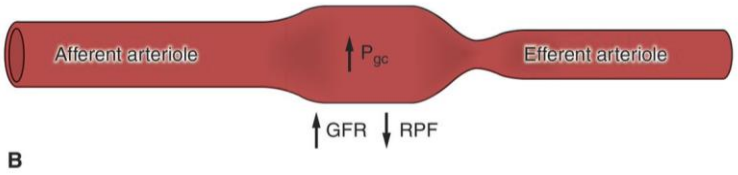
Vasoconstriction of vas afferent leads to ↓ glomerular filtration rate (*the increased Na^+ absorption) and ↓ renal perfusion



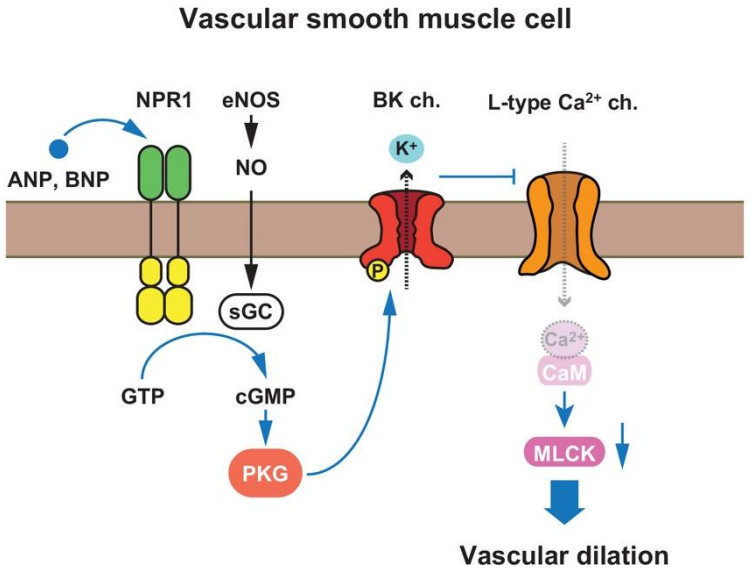
Vasoconstriction of vas efferent with vasodilation of vas afferent leads to ↑ glomerular filtration rate (*decreased Na^+ absorption) and ↓ renal perfusion

*When the GFR spontaneously decreases or increases, the renal tubules and, primarily the proximal tubules, automatically adjust their rate of reabsorption of sodium and water to balance the change in GFR. Thus a constant fraction of filtered sodium and water is reabsorbed from the proximal tubule. This prevents wide fluctuations in sodium and water excretion into the urine and maintains sodium and water balance

ANP: The model of glomerular filtration rate (GFR) and renal perfusion (RPF)



- Natriuretic peptide: Produced by atria and ventricles of the heart with hypertension and increased blood volume; causes vasodilation of afferent arteriole and vasoconstriction of efferent arteriole; modest increase in GFR with little change in RBF



↑ANP, ↑ adenosine, ↑ NO, ↑H^+, ↑lactate and ↓ATP induce <u>hyperpolarization</u> of the cellular membrane of the vascular smooth muscle cells. It occurs via opening K^+ channel that close Ca^{2+} channel and induce the vasodilation

NT-proBNP

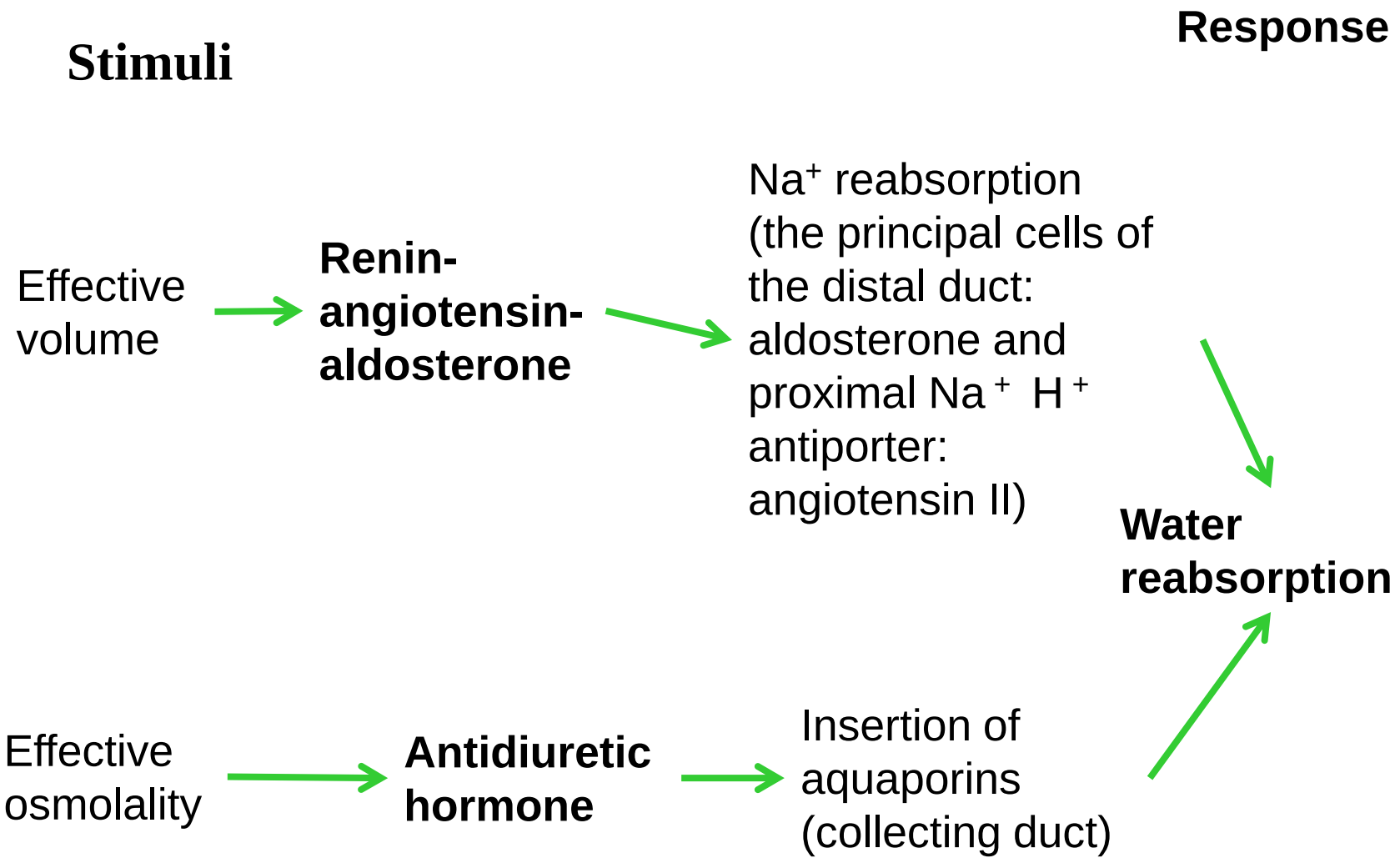
- A hormone-inactive product of the proBNP molecule. It is considered to be a fragment that is incapable of linking to receptors and probably without biological significance.

- It is a peptide with a high validity in cardiologic diagnostics, where it has a clinically comparable statement as its own active hormone BNP.

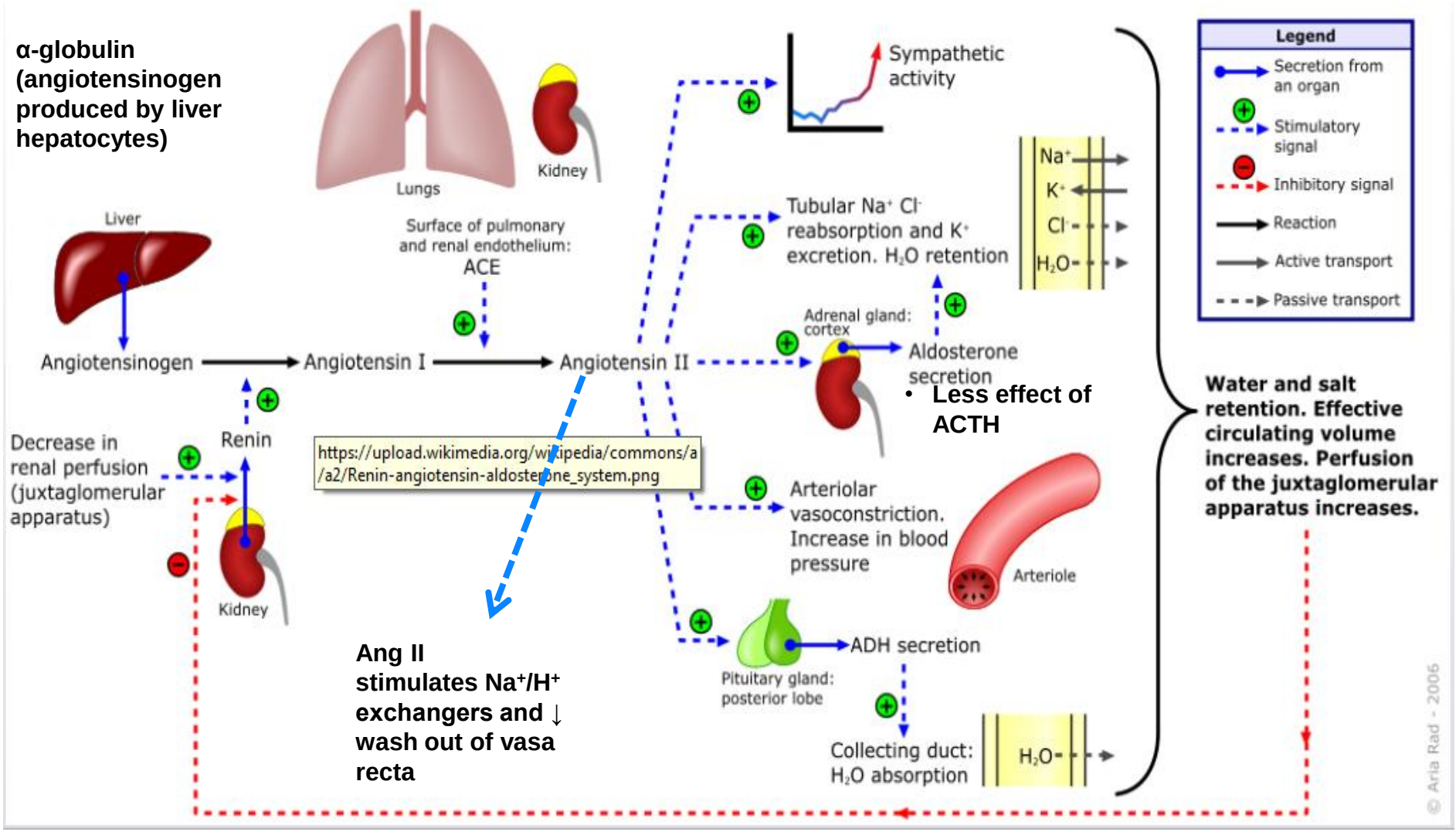
- BNP and NT-proBNP concentrations in plasma grow in people with decompensated heart failure (distension of myocytes heart walls in heart failure as a pump), pulmonary hypertension, renal failure.

- Compared to BNP, NT-proBNP has some advantages such as - stability of molecule, less biological variability, higher degree of standardization).

Water reabsorption



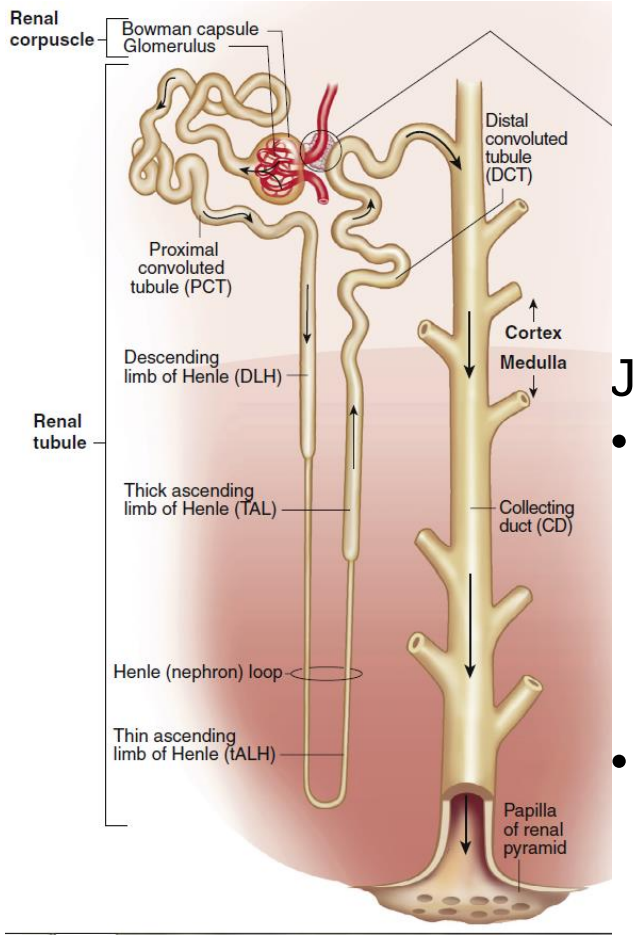
Renin-angiotensin-aldosterone system

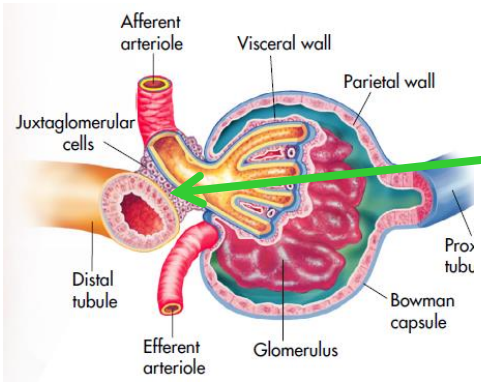


- Vitamin D3 is a potent negative endocrine regulator of renin gene expression
- ACE inhibitors are a class of drugs that reduce blood pressure by inhibiting the formation of angiotensin II and aldosterone

1 Renin production: its production is stimulate by

Renin is an enzyme formed and stored in granular cells of the afferent arterioles of the JGA





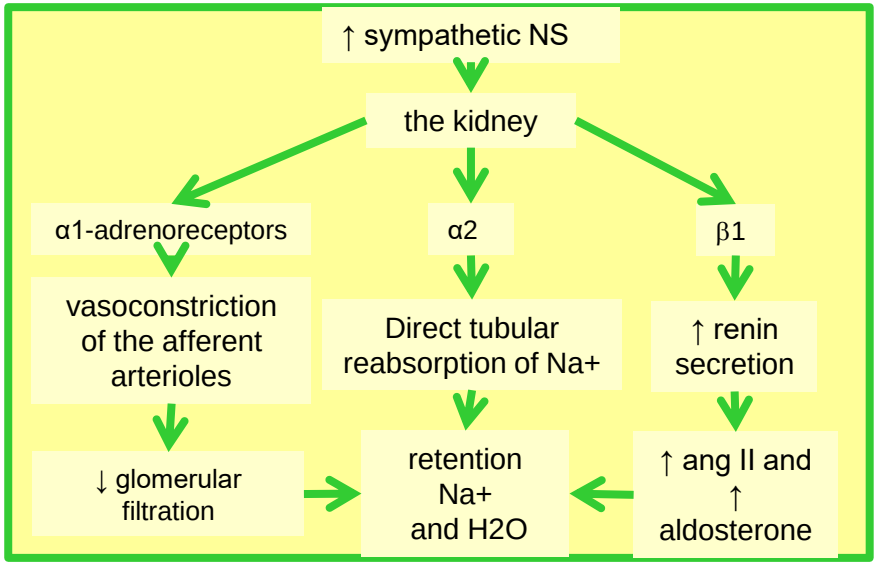
Innervated by SNS (the next slide)

Juxtaglomerular apparatus: 2 parts

- Juxtaglomerular cells are located around the afferent arteriole.
  - Decreased blood pressure (renal perfusion) reduces stretch of the juxtaglomerular cells and trigger renin release.
- Portion of the distal convoluted tubule with specialized Na^+ and Cl^-sensing cells known as the macula densa: ↓Na at the start of distal tubules detected by macula densa increases renin

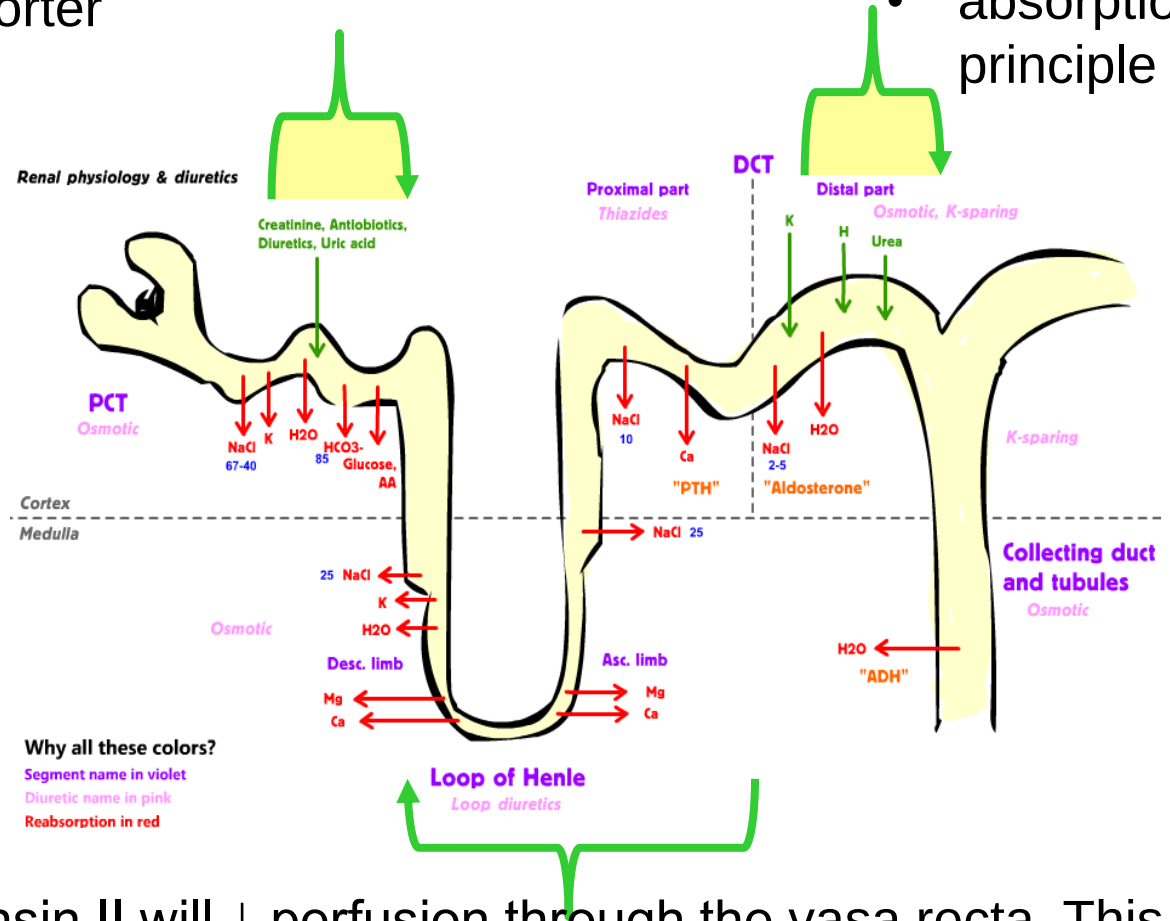
2 Renin production: its production is stimulate by sympathetic nerve system

- Sympathetic nerve via β1 receptors induces renin and via α1 vasoconstriction of vas afferens: low blood pressure detected by baroreceptors in arterial circulation activates sympathicoadrenal system



Role of angiotensin II and aldosterone

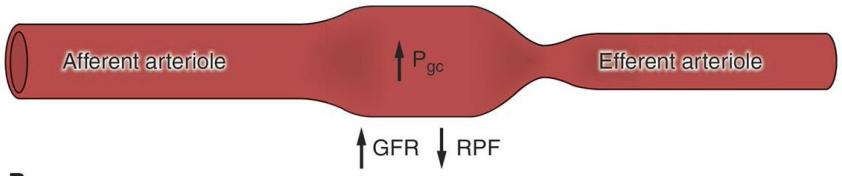
- Angiotensin II stimulates Na^+ H^+ antiporter
- Aldosterone:
  - secretion of K^+ and H^+
  - absorption of Na^+ in principle cells



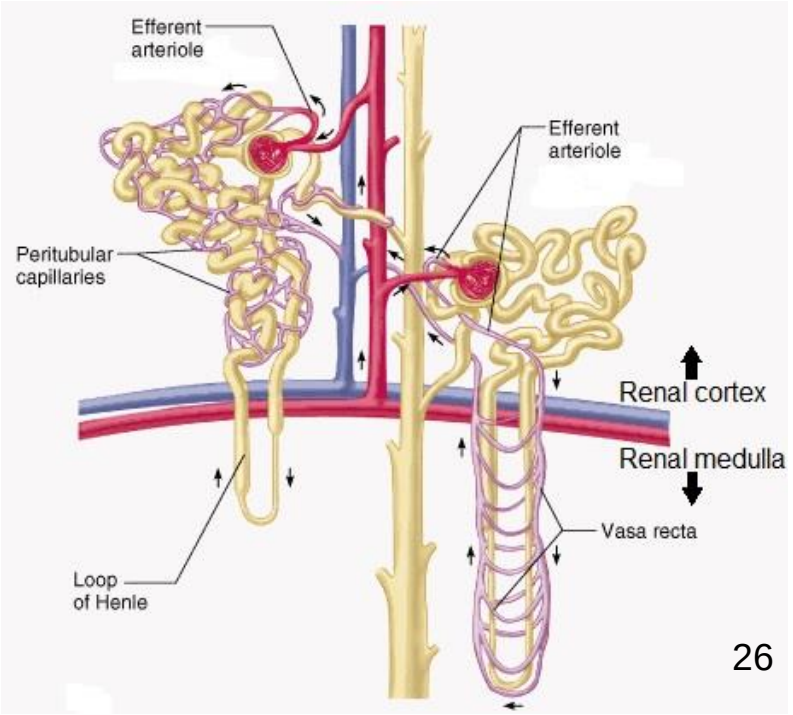
- Angiotensin II will ↓ perfusion through the vasa recta. This decreases the washout of NaCl and urea in the kidney medullary space. It facilitate absorption of tubular fluid that will increase passive reabsorption of sodium along the thick ascending limb of the Loop of Henle.

1 Role of angiotensin II: efferent arteriole

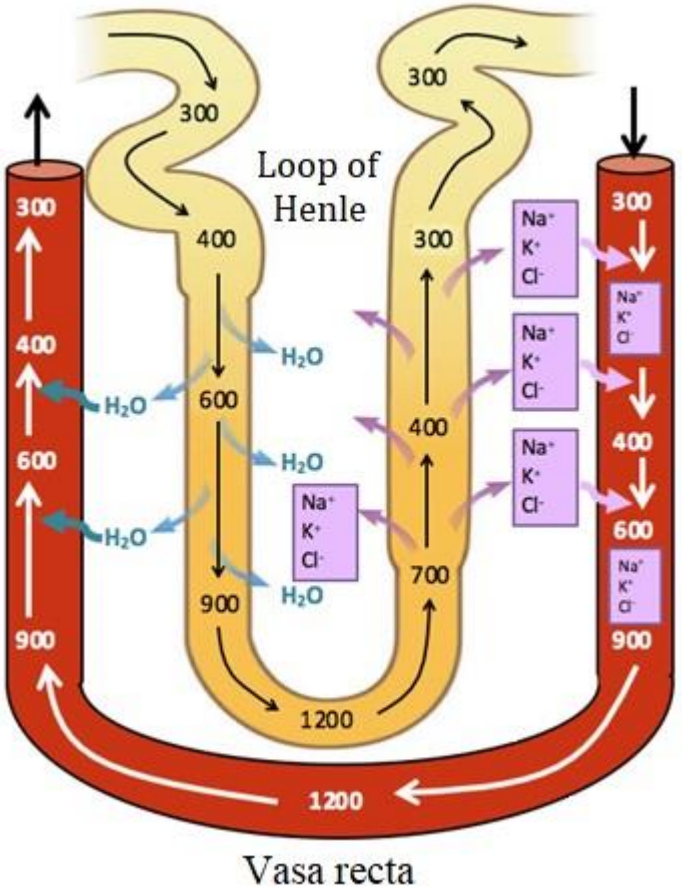
- Angiotensin II. Arteriolar vasoconstriction including the vas efferens (less the vas afferens) will ↑glomerular filtration and decreases renal perfusion in peritubular capillaries. It maintains the GFR even under conditions of renal hypoperfusion, e.g. bleeding. The GF forms primary urine from that Na^+ and H_2O is reabsorbed in the distal tubules.
- In contrast, at higher concentrations or chronic one, angiotensin II induces glomerular remodeling and vascular smooth muscle cells to hypertrophy (hypertension, atherosclerosis, renal disease, and cardiac hypertrophy)



Angiotensin II: mesangial cell contraction

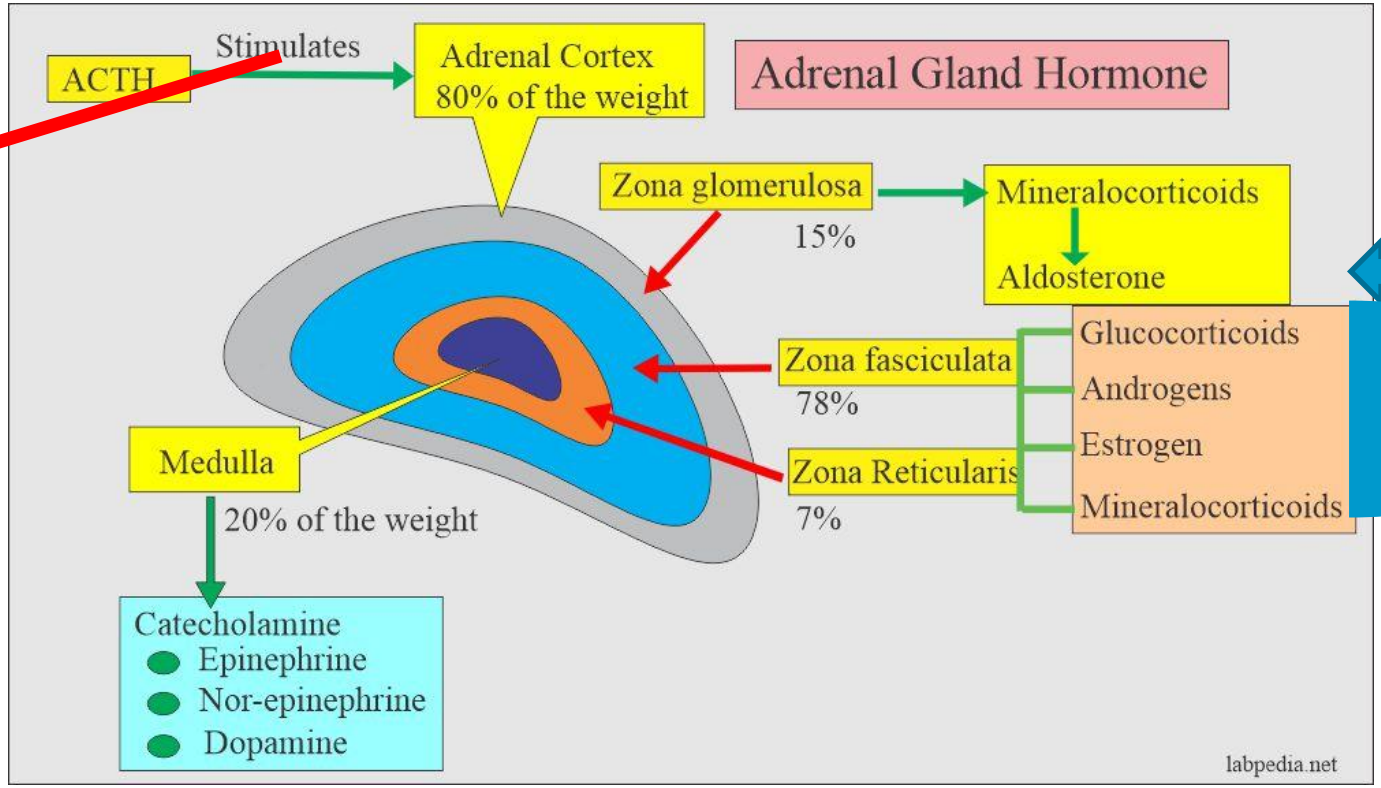


2 Role of angiotensin II: vasa recta



- Physiology: a countercurrent in the loop of Henle and vasa recta
- Angiotensin II – mediated low perfusion through the vasa recta decreases the washout of NaCl and urea from the kidney medullary space. It facilitate absorption of tubular fluid that will increase passive reabsorption of sodium along the thick ascending limb of the Loop of Henle.

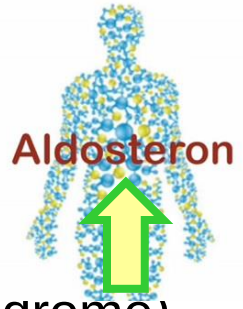
Aldosterone: Adrenal gland hormone



Renin, less ACTH

ACTH

Hyperaldosteronism



Primary aldosteronism (primary hyperaldosteronism)

- It is **increased aldosterone originally**
- Causes:
  - Benign aldosterone-producing adrenal adenoma (Conn's syndrome)
  - Bilateral adrenal nodular hyperplasia: Because cortisol production is low, ACTH concentration increases and causes adrenal hyperplasia.
  - Adrenal carcinomas (rare)

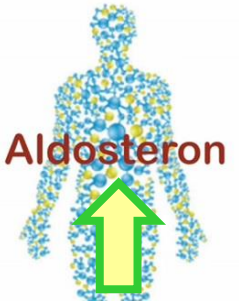
Secondary hyperaldosteronism

- It is **increased renin originally** and secondarily increased aldosterone and renin
- Causes: cirrhosis, ectopic production of renin, coarctation of aorta (narrow aorta), stenosis of renal arteria, decreased cardiac output in cardiac failure (decreased circulating blood volume). During pregnancy, estrogen stimulates the production of renin and angiotensin. In hyperthyroidism, thyroid hormones increase renin.

Pseudohyperaldosteronism

- **low aldosterone**
- activation of the aldosterone receptor without an increase in aldosterone.
  - Cushing's syndrome (high cortisol stimulated receptor for aldosterone)
  - High dose steroid therapy
  - Liddle syndrome, an autosomal dominant genetic disorder

Clinical manifestation of primary hyperaldosteronism



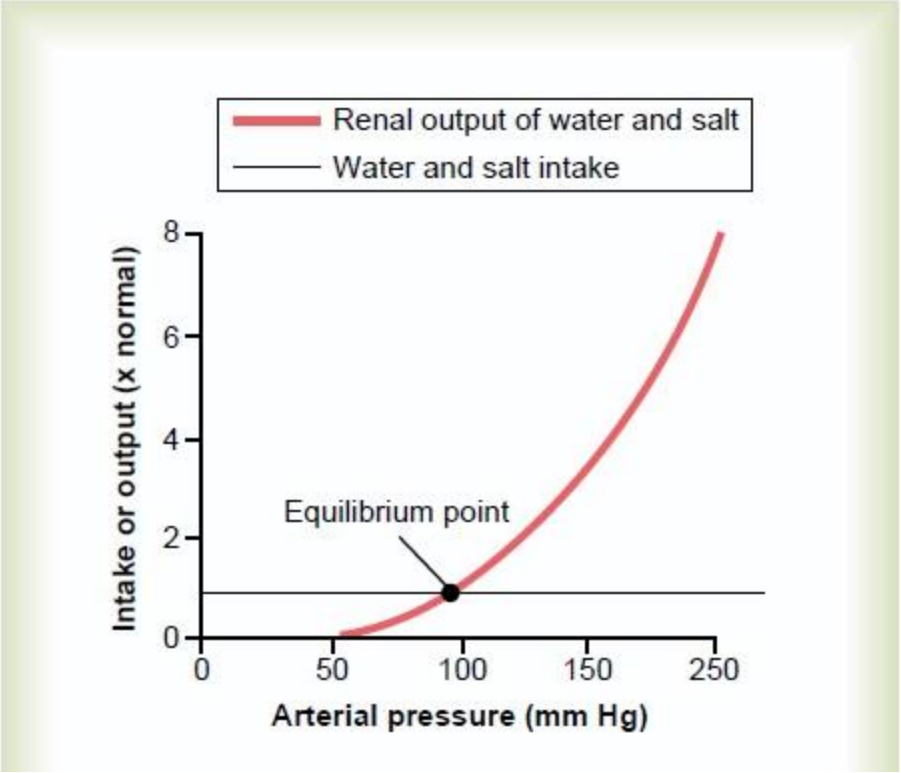
Conn's syndrome :

- Hypertension (↑Na^+ can be diluted and thus normonatremia)
- Renal excretion of potassium (hypokalemia with alkalosis may lead to hypocalcemia)
- No edema: Sodium loss and water loss are maintained by hypervolemia-induced atrial natriuretic factor release, pressure natriuresis, and aldosterone escape (spontaneous diuresis) leading to nocturia and polyuria X secondary hyperaldosteronism is connected with edema.
- The ANP-mediated polyuria can increase urinary losses of potassium (hypokalemic nephropathy with decreased urinary concentration)
- Low renin

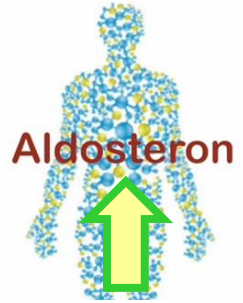
Metabolic syndrome (hypertension, obesity, dyslipidemia, insulin resistance, and hyperglycemia) is associated with primary aldosteronism: Aldosterone reduces insulin sensitivity in skeletal muscle and adipocytes

Pressure-diuresis or pressure-natriuresis

Relatively small increases in glomerular filtration that follow from increases in renal arterial pressure cause relatively large increases in urine and sodium output.

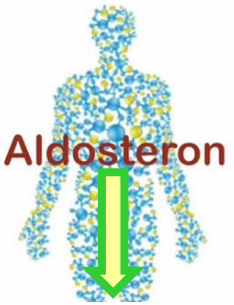


Clinical manifestation of secondary hyperaldosteronism

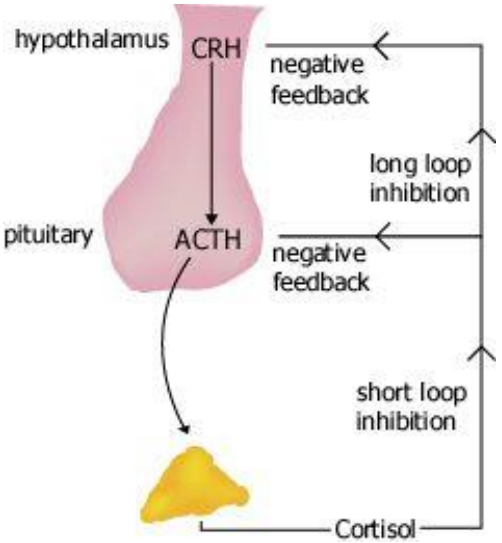


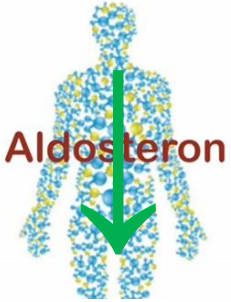
- ✖ Increased renin (stimulated by ↓circulating blood volume)
  - During pregnancy, ↑estrogen stimulates the production of renin and angiotensin.
  - In hyperthyroidism, thyroid hormones increase renin.
- ✖ Hypertension and hypokalemia is less common compared to primary hyperaldosteronism
- ✖ Edema are common

Hypoaldosteronism



- Congenital hypoaldosteronism:
  - Hyperreninemic hypoaldosteronism: genetic defect in enzymes that are important for synthesis of aldosterone
  - Congenital form of Addison's disease (congenital adrenal hypoplasia) or pseudohypoaldosteronism
- Acquired hypoaldosteronism:
  - Primary hypoaldosteronism: Addison's disease (primary adrenal insufficiency, autoimmune): ↓aldosterone, ↓cortisol (primary hypocortisolism) and secondarily ↑ increased adrenocorticotropic hormone (ACTH) and ↑ renin
    - Note: secondary adrenal insufficiency (secondary hypocortisolism, e.g. exogenous glucocorticoids) due to ↓ ACTH does not affect aldosterone but similar to SIADH with ↑ADH as cortisol inhibit ADH
  - Secondary hypoaldosteronism: Acquired hypoaldosteronism: due to low renin: hyporeninemic hypoaldosterinism
  - Pseudohypoaldosteronism: high aldosterone, tubular cells do not response to aldosterone due to renal hypoxia or urine stagnation





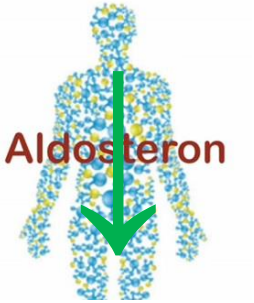
Addison disease: causes



- Autoimmune disease: rare, more common in women
  - 21-Hydroxylase autoantibodies and autoreactive T cells specific to adrenal cortical cells

- Others: infections (tuberculosis, fungal infections, human immunodeficiency virus), infiltrative diseases (amyloidosis, metastatic carcinoma),bilateral adrenal hemorrhage.

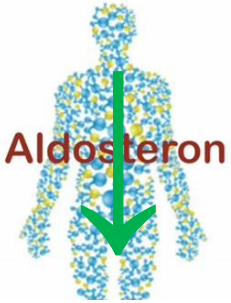
- AD often is associated with other autoimmune diseases: autoimmune polyendocrine syndrome (APS)*:*
  - Type I: autosomal recessive (Addison disease, hypoparathyroidism, mucocutaneous candidiasis)
  - Type II: Addison disease, immune thyroid disease, diabetes mellitus, celiac disease, and hypogonadism.



Addison disease: manifestation I



- ↓ corticosteroid and ↓ mineralocorticoid synthesis and ↑ ACTH
- Serum glucose concentration can be low: decreased gluconeogenesis
- Manifestation of primary hypocortisolism dominate

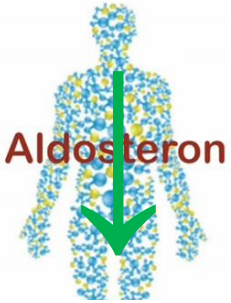


Addison disease: clinical manifestation II



- Weakness and easy fatigability and later anorexia, nausea, vomiting, and diarrhea
- Disturbances in mood and motivation are common

- Hypotension:↓ Na^+ leads to hypovolemia and ↓ glomerular filtration, weakness
- GIT disturbances: celiac disease electrolyte
- Water poisoning: hypovolemia induces ADH that leads to H_2O retention. Water poisoning manifest with headache, nausea, vomiting and glucose fever due to edema in hypothalamus, mainly after administration of glucose.
- Hyperkalemia: both decreased ↓aldosterone-mediated K^+ excretion and ↓ glomerular filtration contribute to ↑K^+ leading to arrhythmia, mild alkalosis or hypochloremic acidosis

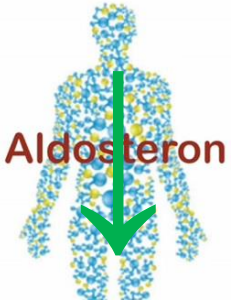


Addison disease: clinical manifestation II



- Hyperpigmentation: ACTH stimulate melanocytes
- Vitiligo (white patchy areas of depigmented skin): autoimmune destruction of melanocytes
- ↓ **adrenal androgen**
  - **In men: the adrenals not a major source of male androgens.**
  - **In women: a loss of some secondary sexual characteristics such as pubic and axillary hair, normally maintained by the adrenal androgens**
- **Acute** Addisonian crisis





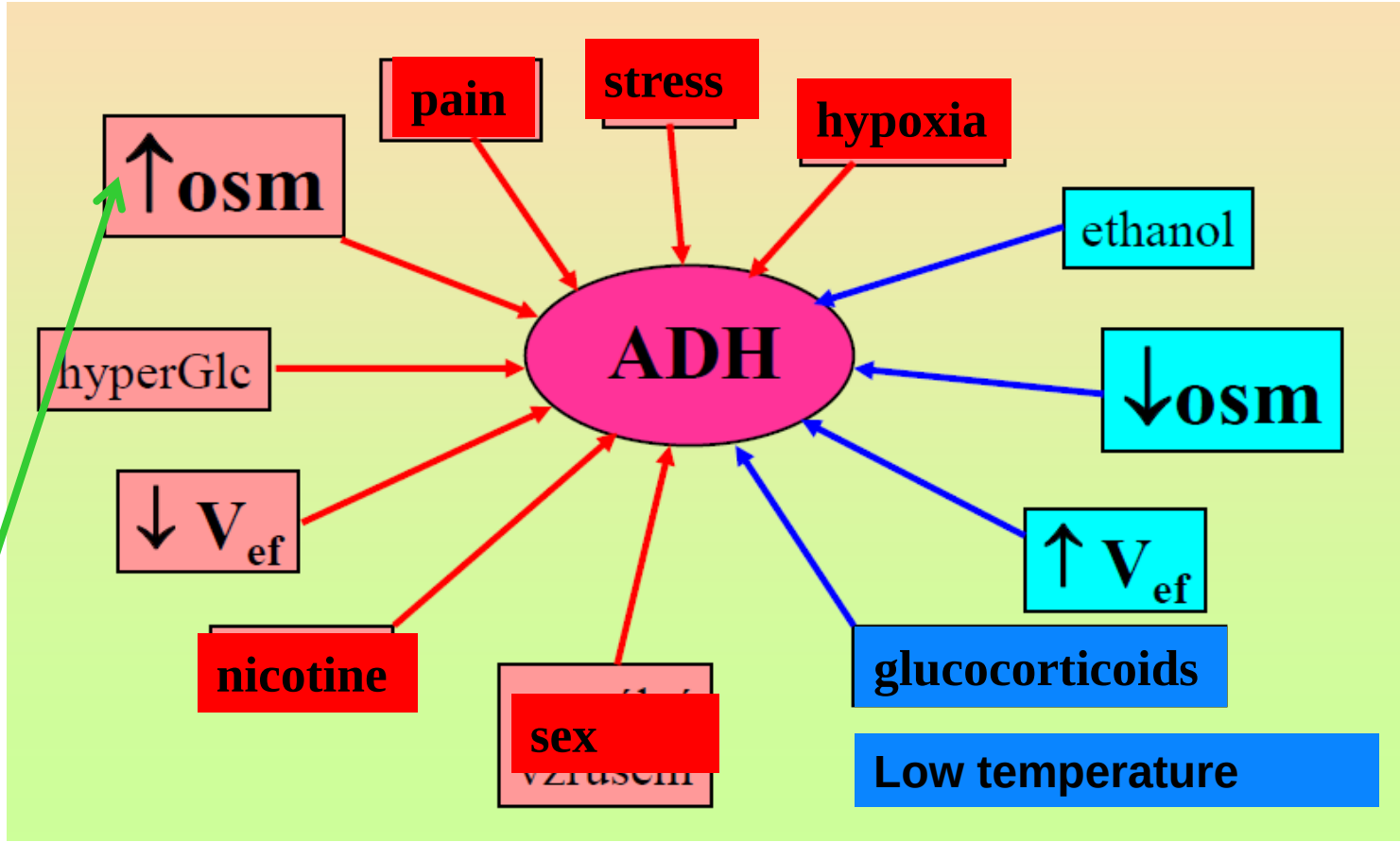
Acute Addisonian crisis

- Sever hypotension and vascular collapse that can lead to circulatory shock
- Combined effect of hypocortisolism, hypoaldosteronism, volume depletion and precipitating factors: infection or ↓ vasomotor tone due to ↓ cortisol
- Acute renal failure: ↑ creatinine and ↑urea
- Eosinophil and lymphocyte counts are often elevated, neutropenia
- ↑fever, confusion, vomiting

Antidiuretic Hormone (ADH) or vasopressor

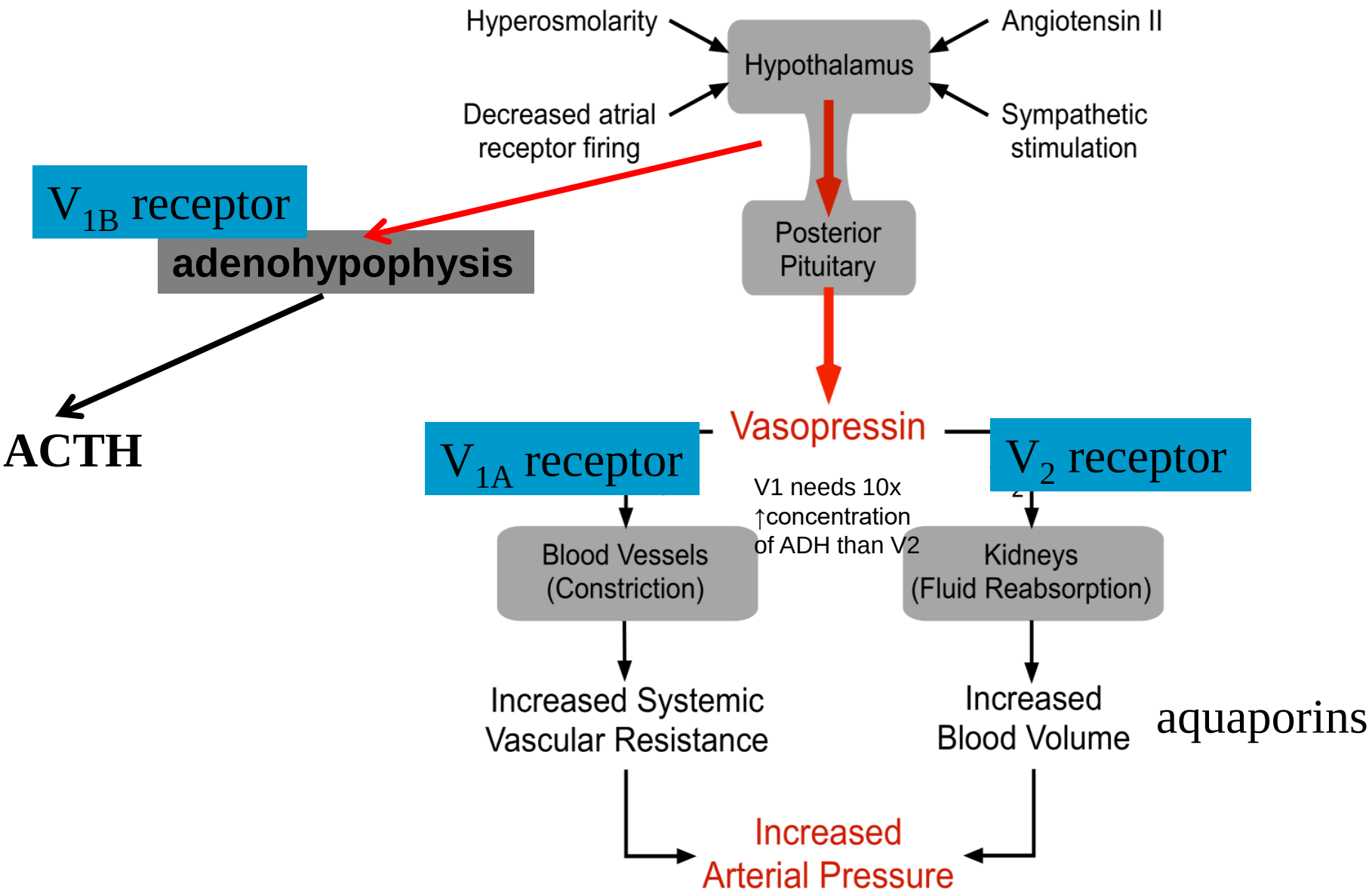
The secretion of ADH is stimulated by

The secretion of ADH is inhibited by



- Induced primarily by the osmoreceptors in the hypothalamus.
- Induced less by hypovolemia: e.g. In circulatory shock, an arterial hypotension induces ADH that inactivate K^+ channel leading to vasoconstriction but later ADH is run out and vasodilation
- Induces primarily water retention

Antidiuretic Hormone (ADH) or vasopressor



Antidiuretic Hormone (ADH): conditions

- Low function of antidiuretic hormone: **Diabetes Insipidus**
- Increased function of antidiuretic hormone: **Syndrome of Inappropriate Antidiuretic Hormone (SIADH or Schwartz-Bartter syndrome)**

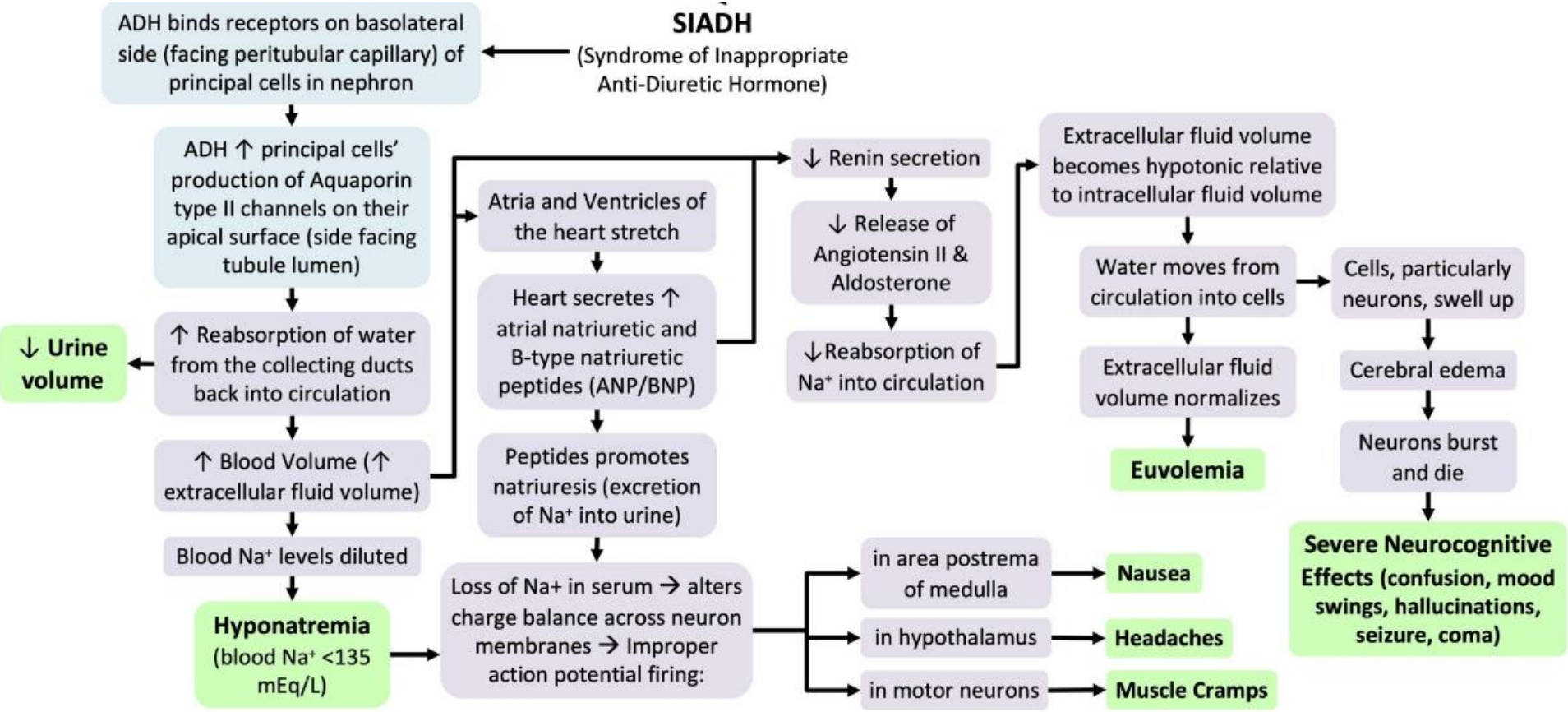
Syndrome of Inappropriate Antidiuretic Hormone (SIADH or Schwartz-Bartter syndrome)

- Increased function of antidiuretic hormone
- Causes:
  - CNS disorders, e.g. stroke, hemorrhage, infection or tumor
  - Ectopic production of ADH by a tumor
  - Pulmonary disorders (unknown mechanism)
- Mechanism: ↑ADH induces water retention leading to dilutional hyponatremia
  - **Absolute urine osmolarity is low but ratio between urine and plasma osmolarity is ↑.** (Urinal Na^+ is concentrated due to poor water retention but not too much due to hyponatremia.)
  - edema↑ does not have to be present due to compensatory ANP and diuresis.

- Manifestation: lethargy, myoclonus, asterixis (flapping tremor , temporary loss of motor tone) and confusion or coma

- https://www.youtube.com/watch?v=sEnp2ss8VoA

Syndrome of Inappropriate Antidiuretic Hormone (SIADH or Schwartz-Bartter syndrome): mechanism



Signs and symptoms depend upon the rate (chronic vs acute: tolerated better/less) and severity of hyponatremia:

- Acute or fast decline of Na^+: cerebral edema
- Chronic: adaptive responses counteracting swelling in brain cells and is traditionally defined as an asymptomatic

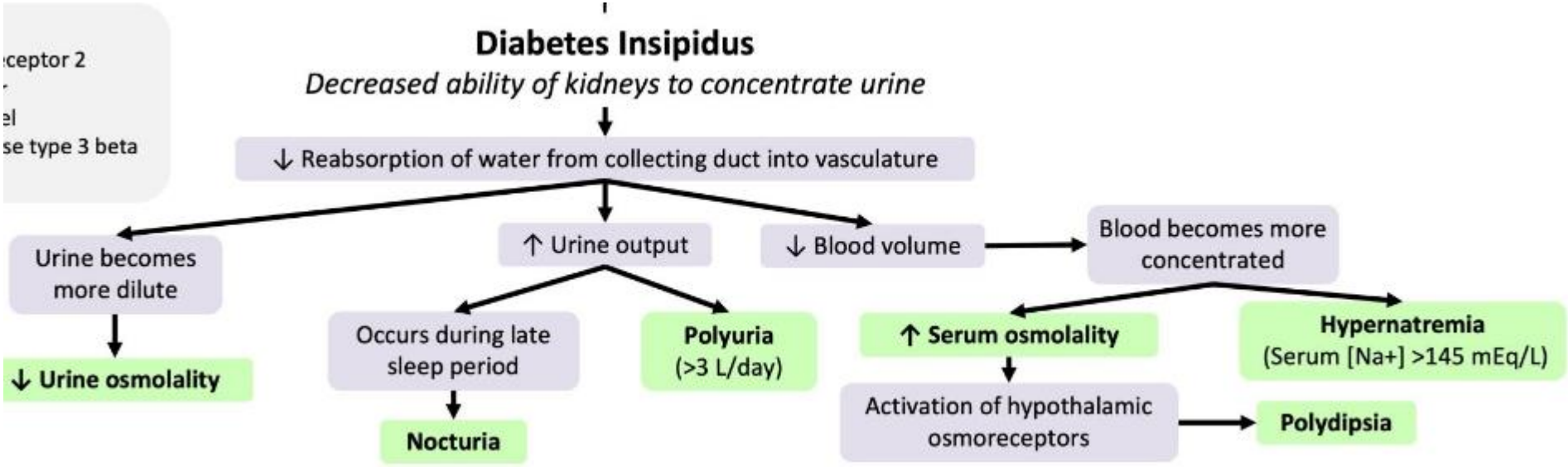
Diabetes Insipidus

- Low function of antidiuretic hormone and loss of hypotonic water (increased urination) leads to hypernatremia that induces thirst

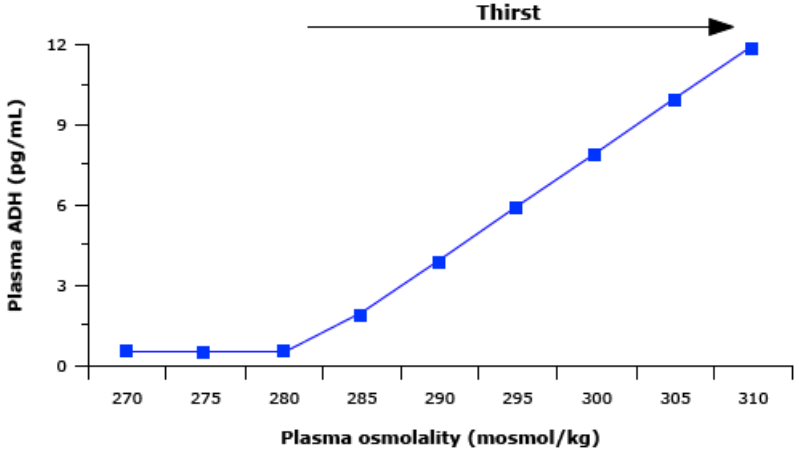
  - Central (neurogenic) (low level of ADH)
    - causes: pituitary dysfunction due to ischemia or hamorrhage, pregnancy (placenta can produce the enzymes that degrade ADH)
    - Synthetic ADH will help

  - Peripheral (nephrogenic) (normal level of ADH)
    - causes: receptor dysfunction due to genetic mutation in V_2 receptor
    - Synthetic ADH will not help. Vasoconstriction (V_{1A} receptor) is not affected as V_2 receptor has defect.

Diabetes Insipidus: mechanism



Signs and symptoms depend upon the rate (chronic vs acute: tolerated better/less) and severity of hypernatremia:
hypernatremia can suck out water from brain and cause bleeding



Hypovolemic and hypervolemic states

Hypovolemic and hypervolemic states: content

- Dehydration
  - Symptoms
  - Examples: isotonic, hypotonic and hypertonic
- Hyperhydration: symptoms
  - Symptoms
  - Examples: isotonic, hypotonic and hypertonic

Dehydration: symptoms

- Headache and thirst
- Weight loss
- Decreased urine output (oliguria), and concentrated urine (with the exception of diabetes insipidus and others).
- Skin turgor may be normal or decreased. Dry skin and mucous membranes.
- Symptoms of hypovolemia: tachycardia, weak pulses and (postural) hypotension (peripheral vasoconstriction)
- Others: Elevated levels of hematocrit (with the exception of bleeding),serum sodium concentration, urine and serum osmolality, and urine specific gravity are associated with moderate water loss.

Dehydration: hematocrit

- The **hematocrit** (**Ht** or **HCT**), the volume percentage (vol%) of red blood cells (RBC) in blood.

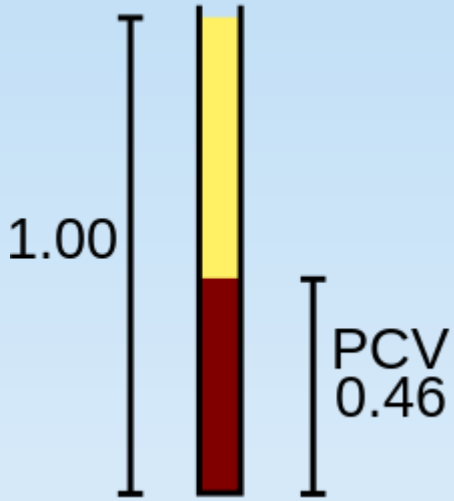


Diagram of packed cell volume


Poor skin turgor

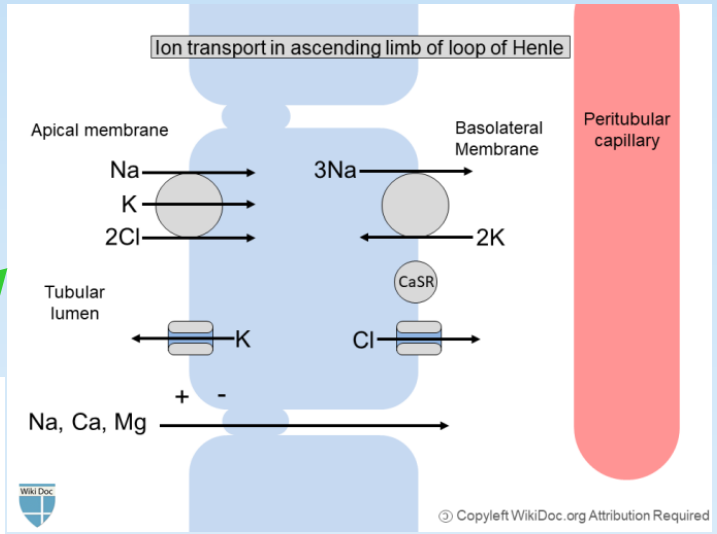
Dehydration: forms

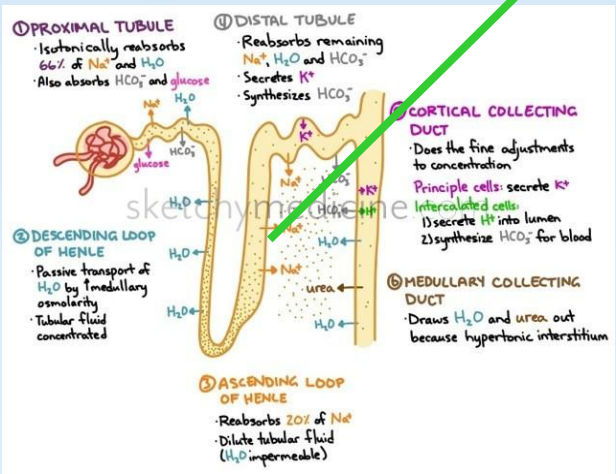
Dehydration	**Increase hematocrit in isotonic and hypotonic (not in plasma loss)**	
Isotonic (Normal Na^+)	Hemorrhage, profuse bleeding, burn	Loss of isotonic fluid: hypovolemia with peripheral vasoconstriction, tachycardia and RAAS, ADH activation. Note Despite high aldosterone, ↑K^+ excretion does not have to be present due to low urine in the distal tubules that will concentrate K^+. By contrast, loop diuretics will increase urine in the distal tubules and K^+ excretion)
	Puncture of ascites, sever wound drainage after operation.	
	Fluid leakage to the 3th space: paralytic ileus, pancreatitis	
Hypotonic (Decreased Na^+)	Low aldosterone	Loss of water is less than Na^+ loss: cellular edema
	nephropathia loosing salt: if primary renal disease affected the renal marrow osmolarity e.g. Pyelonephritis (bacterial infection of renal pelvis)	
	Hypotonic saline or pure water replacement after endurance exercise (No issue as healthy kidney will compensate: reabsorbed Na^+)	
	Bartter syndrome	
	Overdose of loop diuretics	
Hypertonic (Increased Na^+)	Profuse sweating, diarrhea (sweat, stomach fluid and stool is hypoosmotic, sever diarrhea has isoosmotic fluid)	Loss of water is higher than loss of Na^+: cellular exsiccation
	Polyuria at acute renal failure (the remaining nephrons cannot reabsorb Na+ and ↑ water loss due to osmotic diuresis with hypoosmotic urine)	
	low antidiuretic hormone (diabetes insipidus, thirst due to ↑ osmolarity)	
	osmotic diuresis in diabetes mellitus	

Barter syndrome (BS)

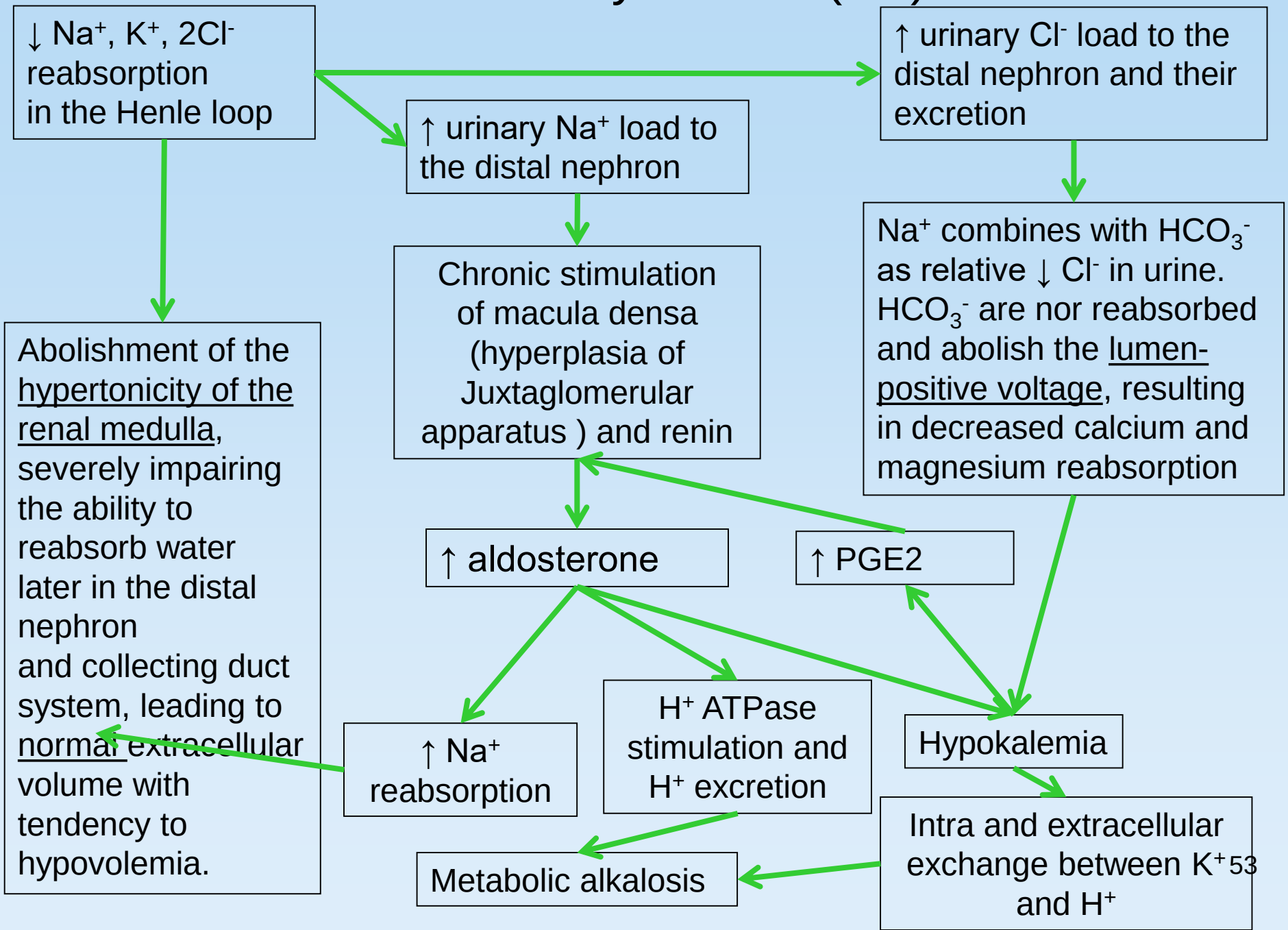
mutations involving the Na^+-K^+-$2Cl^-$ cotransporter in the thick ascending limb of the loop of Henle. Na^+-K^+-$2Cl^-$ cotransport one sodium, one potassium, and two chloride ions across the apical membrane of the tubule.

- Primary hypereninism and secondary hyperaldosteronism
- Hypokalemia
- Hypochloremia
- Metabolic alkalosis
- Tendency to hypovolemia





Bartter syndrome (BS)



Hyperhydration: forms

Hyperhydration	decrease hematocrit in isotonic and hypertonic	
Isotonic (Normal Na^+)	Accumulation of isotonic fluid due to↑ hydrostatic or ↓ oncotic pressure=edema with isotonic fluid accumulation (secondary not primary ↑aldosterone)	severe congestive heart failure
		cirrhosis or hepatic failure
		nephrotic syndrome, >3,5 g/24 hours
	Overdose by anti-inflammatory drugs that ↓ PGE2	PGE acts via vasodilation in vas afferent
Hypotonic (Decreased Na^+)	Water is accumulated more than salt: Increased mean erythrocyte volume (cellular edema)	↑antidiuretic hormone (SIADH) (hypervolemia, no edema) hyponatremia and hypokalemia °
		psychogenic disease (compulsive water drinking)
		Acute renal failure with oliguria/anuria (catabolism)
Hypertonic (Increased Na^+)	Salt is accumulated more than water: Decreased mean erythrocyte volume (cellular exsiccation)	↑aldosterone (Conn‘s syndrome) (hypervolemia, no edema)~
		salt water
		Acute renal failure

°Primary effect on aquaporins: accumulation of water and diluted Na+. Euvolemic hyponatremia due to secondary induced ANP.
~Due to water retention, hypernatremia does not have to be present. No edema: hypervolemia induces atrial natriuretic factor and diuresis

Acute renal failure with anuria or oliguria

- Decreased glomerular filtration
  - ↑ serum urea to creatinine ratio:
    - Normally urea is reabsorbed: The low filtration of urea due to low GF will increase urea reabsorption.
    - Normally creatinine is secreted by tubular cells if serum creatinine is high. The low filtration of creatinine will increase serum creatinine but serum creatinine is secreted by tubular cells.
- Renal tubules are not damaged:
  - ↑ urinal osmolarity due to urea (Na+ is low)
  - Na+ is reabsorbed by RAAS leading to water retention: life threatening edema (pulmonary edema), heart failure, cell swollen, high intracranial pressure
    - Catabolic state of metabolisms is additional source of water that is not excreted: hypoosmolar hyperhydration (hypervolemia)
  - Oliguria
- Metabolic acidosis: accumulation of waste product of metabolism and low excretion of H+
- Hyperkalemia: in the last stage, in early stage: K+ usually excreted by tubules (arrhythmia)

Normovolemia: forms

Normovolemia		
Isotonic (Normal Na^+)	normal	
Hypotonic (Decreased Na^+)	hypotonic saline or pure water replacement after endurance exercise	Loss of water is less than Na^+ loss: cellular edema
Hypertonic (Increased osmotically active molecules)	hyperglycemia	Loss of water is higher than loss of osmotically active molecules: cellular exsiccation
	Accumulation of ketone bodies in diabetes	

References

Look up the particular theme in the following indexes

- Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014.
- McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018.
- Silbernagl S, Lang F. Color Atlas of Pathophysiology, 3rd Ed. Thieme, 2016.
- https://pfyziol.upol.cz/ - selected text passages (according to teachers' recommendations)